

# Hyunwoo Yang

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## EDUCATION

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DigiPen Institute of Technology | Redmond, Washington

B.S. in Computer Science | GPA: 3.913

Graduate 2028

Relevant Courses: Computer Graphics, Advanced C/C++, Game Projects, Data Structure and Algorithms

## TECHNICAL SKILLS

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Programming Languages: C/C++, C#, Java Script, GLSL, Rust

Others: Unity, Unreal, Vulkan, OpenGL, Linux, Github, Perforce

Soft Skills : Teamwork and Collaboration, Solving Problem, Creativity, Communication, Growth Mindset

## PROJECTS

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### [LostGuns](#) | Solo Develop

01 / 2026 – 03 / 2026

- A 2D multiplayer shooter recreating the core mechanics of the mobile game LostGuns in Unity, built from scratch by studying gameplay footage.
- Reverse-engineered the original's systems (movement, jumping, ladder traversal, and weapon switching) and added friend-to-friend online play via Unity Relay.
- Integrated acquired 2D assets by rigging them with DragonBones, and manually authored skeletal animations to faithfully replicate the original game's visual dynamics.
- Implemented server-authoritative hit detection with lag compensation, rewinding player positions to account for RTT and interpolation delay, clamped to a 0.4s maximum.

### I.G.Y | TechLead

09 / 2025 – 06 / 2026

- Isometric city-builder simulation inspired by Caesar 3 and SimCity 2000, built on a custom C++/OpenGL engine.
- Led technical architecture for a team of 5 across the engine, simulation, and manage the git.
- Expanded my Computer Graphics coursework renderer into a standalone engine, Dokev, wrapping raw OpenGL into single-function shape, color, and texture calls so teammates new to graphics could focus on gameplay and AI.
- Engineered the tick-based simulation core (economy, resource flow, population) with A\* citizen pathfinding, optimized with spatial partitioning to sustain hundreds of agents.

### Help me! Doctor Water | QA

03 / 2023

- 2D vertical-scrolling escape-action game; Dr. Water ascends an erupting volcano, fighting enemies and bosses with a 5-weapon arsenal
- Built enemy attack behavior (projectile spreads, charges, wave spawns) and the player's 5 weapons, each with distinct damage, fire rate, and projectile pattern.
- Tested all stages and bosses as QA through systematic playtesting, logging bugs in enemy behavior, weapon interactions, and progression.
- Mircrodegree Expo Best Award , KMU, 2023

## EXPERIENCE

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### Daegu Futsal Team | Captain | Daegu, South Korea

03 / 2025 – 07 / 2026

- Captain of a futsal team based in Daegu, South Korea.

### Volunteer | Fundraising Volunteer | Daegu, South Korea

02 / 2025 – 08 / 2025

- Collaborated with World Vision, habitat to provide aid to vulnerable populations.
- Fostered community engagement to support fundraising initiatives.

### Military Service | Leader | Daegu, South Korea

08 / 2023 – 02 / 2025

- Completed mandatory military service with an honorable discharge at the rank of Sergeant
- Led a mortar unit squad, responsible for training and coordinating a team of soldiers during operations.

